Introduction to Data Feminism

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Digital Feminisms
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Northeastern University
NULab for Digital Humanities and
Computational Social Science

Feel free to ask questions at any point during the presentation!

Workshop Agenda:

- What is data?
  - Types of Data
  - Physical Technology of Digital Data
- Data Feminism
  - Why Does Data Science Need Feminism?
  - Principles of Data Feminism (D'Ignazio and Klein 2020, 2024)
- Discussion: Preparation for Next Class

All materials can be accessed here: bit.ly/sp26-savit-dataethics

What is Data?

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Discussion: What is Data?

- According to D'Ignazio and Klein (2020, p. 17) “Many people think of data as numbers alone, but data can also consist of words or stories, colors or sounds, or any type of information that is systematically collected, organized, and analyzed.”
- What other types of data have you noticed in your daily life?

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Types of Data

Image source: uxcel

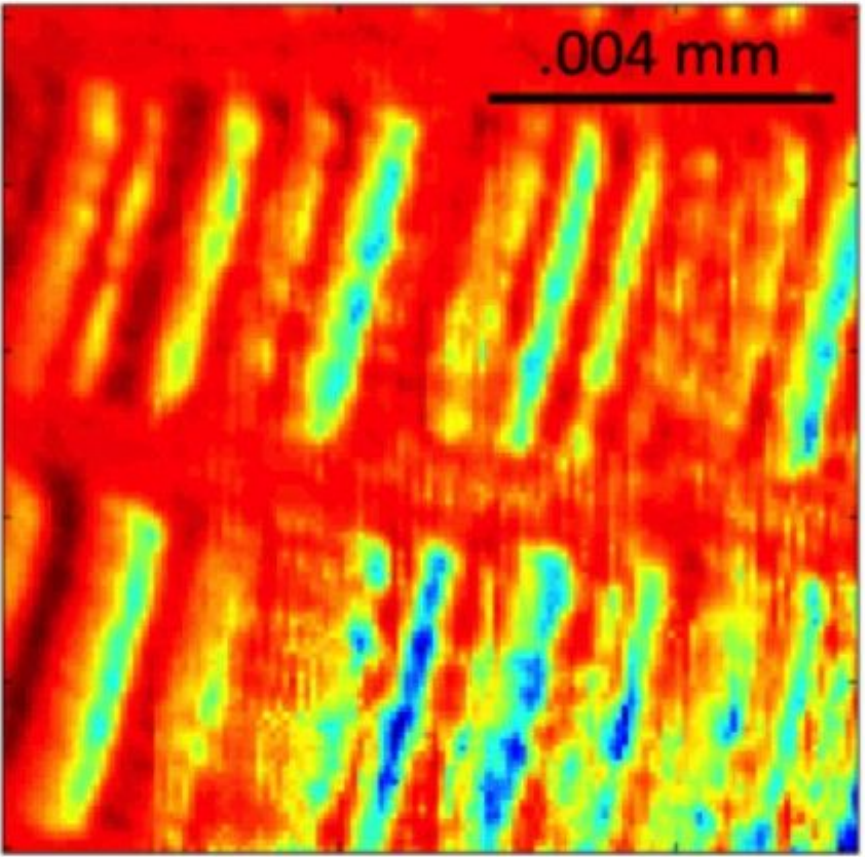
- Quantitative data: Numerical information, such as “1,000 bees” or “Ten books sold”
- Qualitative data: Non-numerical information such as interview transcripts, book text, or images
- Metadata: Data that provides information about or describes data, like the fields “author,” “title,” and “place of publication” for a book.
- Big data: Large and continually growing datasets, often gathered through digital technology

Physical Technology of Digital Data 1/2

- Computers store data by converting all information—text, images, or audio—into binary code (0s and 1s). (Wikipedia)
- Computers need to read from "storage" when accessing data and write to it when saving.
- The picture shows a hard drive which is a common storage device for digital data.

Image source: Wikimedia Commons

Physical Technology of Digital Data 2/2



Close up of the 0s and 1s in the hard drive. The 0s are blue, and the 1s are red.

Berger (2019). [Here's What the Data on Your Hard Drive Looks Like]. *Discover*.

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Cloud storage and data centers

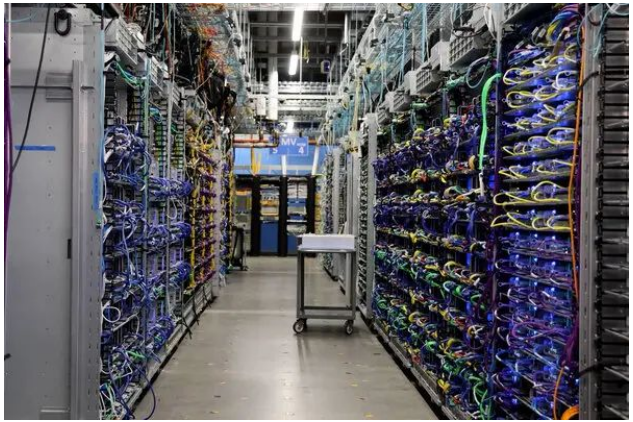

Image source: The New York Times

- "Cloud" storage lets you store data on servers (instead of on your computer) that are accessed via the Internet (AWS, "What is Cloud Storage"). An example of this is Google Drive. Despite the naming, this data storage is still grounded in physical devices.
- Servers and other computing technologies are housed in data centers. These technologies support capabilities such as cloud storage and artificial intelligence (Wikipedia).

What is Data Feminism?

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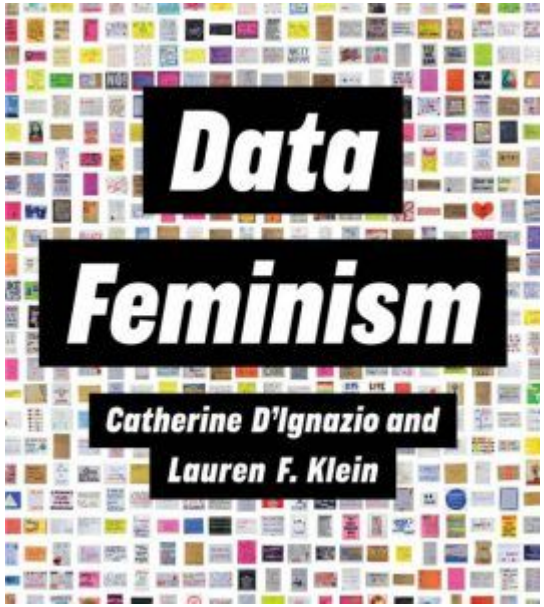
Data Feminism

- In *Data Feminism* (2020), Catherine D'Ignazio and Lauren F. Klein apply intersectional feminism to data ethics and data science to understand how power relations are shaped by these fields
- They argue that data science has been used to support discriminatory power structures and show how data science can be used differently to promote justice



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Why does data science need feminism?

- “Privilege hazard” is the typically unintentional exclusion of those who do not have privileged identities (e.g. are not white and male) from decision making (D'Ignazio and Klein 2020, p. 28).
- Having multiple diverse perspectives improves decision making (D'Ignazio and Klein 2020, p. 28).

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The White Men Default

What is life like for a “straight, white, cisgender man with formal technical credentials who lives in the United States” (D'Ignazio and Klein 2020, p. 28 - 29)?

- Ease of home purchasing and credit card applications
- Social approval of relationships
- No need for workplace accommodations after child birth
- “[T]he world...has been designed for people just like him” (D'Ignazio and Klein 2020, p. 28-29).

Data Feminism explores how the world can be designed for everyone else too.

Data Injustice: Healthcare

- Health care data used for data analysis is often biased by existing injustices, this can further discrimination (Nature).
- Caucasians make up 80 percent of collected data in the fields of genomics and genetics, making findings less relevant to other groups (Harvard School of Public Health).
- "Black women are over 3 times more likely than white women to die from pregnancy- or childbirth-related causes" (CDC 2017, as cited in p. 22, D'Ignazio and Klein 2020; See the Hear Her Campaign).

Goal of Data Feminism

"The work of data feminism is first to **tune into how standard practices in data science serve to reinforce these existing inequalities** and second to **use data science to challenge and change the distribution of power.**" (D'Ignazio and Klein 2020, p10).

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Principles of Data Feminism

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Seven Principles of Data Feminism by D'Ignazio and Klein 2020 (1/3)

- Principle 1: Examine power.
  - The first step is to name and explain the forces of oppression embodied in our lives and data that are often normalized.
- Principle 2: Challenge power.
  - Mobilize data science to challenge the existing unequal structure and to work towards justice. Actions include collecting counterdata, analyzing, imagining “co-liberation,” and teaching (p. 53).

Seven Principles of Data Feminism by D'Ignazio and Klein 2020 (2/3)

- Principle 3: Elevate Emotion and Embodiment.
  - Using emotional visualization might be more effective than bar charts.
- Principle 4: Rethink Binaries and Hierarchies.
  - “Challenge the gender binary, along with other systems of counting and classification that perpetuate oppression” (p. 97).

Seven Principles of Data Feminism by D'Ignazio and Klein 2020 (3/3)

- Principle 5: Embrace Pluralism
  - "[The] most complete knowledge comes from synthesizing multiple perspectives, with priority given to local, Indigenous, and experiential ways of knowing." (p. 125)
- Principle 6: Consider Context
  - "data are not neutral or objective. They are the products of unequal social relations, and this context is essential for conducting accurate, ethical analysis." (p. 149)
- Principle 7: Make Labor Visible
  - Data science requires much work, which should be visible and valued.

Example of Emotional Visualization

"Do Women Have to Be Naked to Get into the Met. Museum? An infographic (of a sort) created by the Guerrilla Girls in 1989, intended to be displayed on a bus billboard. Courtesy of the Guerrilla Girls." (p. 86, D'Ignazio and Klein 2020)

Example of Challenging Gender Binary

- The Digital Transgender Archive is a project dedicated to making transgender history accessible and digitized.

Smith, Bobby. "Performer Exiting a Stage." Photograph. 1950. *Digital Transgender Archive*, https://www.digitaltransgenderarchive.net/files/p8418n32c (accessed November 25, 2024).

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Principles for AI by Klein and D'Ignazio 2024 (1/3)

- In a 2024 article, "Data Feminism for AI", Lauren Klein and Catherine D'Ignazio identify two additional principles focused on AI
- These pertain to environmental impact and consent

Principles for AI by Klein and D'Ignazio 2024 (2/3)

- Principle 8: Environmental Impact.
  - Data centers and computers that runs generative AI models take tons of energy and use a lot of water to cool down. One should be mindful of environment impact when using AI.
  - Google used 5.6 billion gallons of water in 2023 (p. 108).
  - An average Meta data center consumes as much electricity as 150,000 average homes (p. 108).

Principles for AI by Klein and D'Ignazio 2024 (3/3)

- Principle 9: Consent.
  - Training data used for AI modeling, e.g. social media posts, might be obtained without consent of the users.
  - "AI is currently facilitating the mass, non-consensual exploitation of pornographic images of women in the form of deep fakes." (p. 108)
  - Data feminists should build systems that values consent as its core.

Analyze with Data Feminist Lens

Prompt: Write two sentences about the greatest discoveries in the social sciences and humanities and who made those discoveries. Model: Claude, Haiku 4.5, Prompted on 11/25/25.

The discovery of the unconscious mind by Sigmund Freud fundamentally transformed psychology and our understanding of human behavior, revealing that much of what drives us operates outside our conscious awareness. Max Weber's analysis of the relationship between Protestant ethics and capitalism reshaped sociology by demonstrating how cultural and religious values shape economic systems and social structures in ways that transcend simple material determinism.

What is the power dynamic in this example? How would you challenge it?

Example Analysis

Examine Power	The two “greatest” discoveries are done by white men. Who made such evaluation?
Challenge Power	What would be the answers if we specify “women” scholars or scholars from the global south?
Make Labor Visible	AI companies use labor from low income countries to label harmful contents to improve LLM models. This could cause harm to them and the AI company did not acknowledge their labor.
Environmental Impact	Generative AI models consume massive amount of energy to run and the computers take a lot of water to cool down. Consider using less computational-heavy methods before using Gen AI.

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Discussion: Preparation for Next Class

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Discussion

- How can data feminism help us make better decisions with our data?
- How can we apply the principles of data feminism to our own work?

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For Further Exploration

Copyright and fair use handout

Data Ethics handout

Data Privacy handout

Thank you!

Developed by: Sara Morrell, Zhen Guo, Emily Sullivan, Claire Lavarreda, Shannon Peifer

- For more information on DITI, please see: https://bit.ly/diti-about
- Schedule an appointment with us! https://bit.ly/diti-meeting
- If you have any questions, contact us at: nulab.info@gmail.com
- We'd love your feedback! Please fill out a short survey here: https://bit.ly/diti-feedback

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